Swarm (or Swarm *mode*)—Docker's homegrown solution—has been built into the Docker Engine since version 1.12, so you already have it installed. Although it took some time to become production-ready, Swarm is now a mature, capable, low-ceremony orchestrator. While it's still missing one or two desirable features (such as autoscaling), it's a well-thought-out piece of software and puts a lot of power at your fingertips. You'll get to experience Swarm first-hand in the upcoming chapters.

The other big player is Kubernetes, which is an open source tool created by Google, but it is now under the care of the Cloud Native Computing Foundation (CNCF).[1] Not only does Kubernetes support autoscaling, but it also provides more control over how your apps are architected. However, this rich feature set and expressiveness comes at a cost: it is considerably more complex than Swarm, takes more effort to learn, and its config files are more verbose. Additionally, installing Kubernetes by hand can be a lengthy and complicated task itself.

Kubernetes and Swarm have a similar high-level architecture. Both distinguish between worker nodes that run containers and manager nodes that manage the workers and orchestrate containers on them. One notable difference is that Swarm manager nodes can run workload containers *in addition* to their cluster management and orchestration role, whereas Kubernetes manager nodes can't. Also of note, Kubernetes currently uses offensive master/slave terminology to refer to manager and worker nodes, though I am hopeful this will soon be changed.[2]

When learning either Swarm or Kubernetes, getting to grips with their conceptual models[3,4] is key. Although there are similarities, each requires some getting used to. For example, Swarm has the concept of a stack, which envisions the application as a group of underlying services. Kubernetes, on the other hand, breaks this down further into a Deployment, made up of ReplicaSets, which in turn are made up of Pods.

Here's a very high-level summary of the two:

	Swarm	Kubernetes
Open source	Yes	Yes
Created by	Docker	Google

1. https://www.cncf.io
2. https://github.com/kubernetes/website/issues/6525
3. https://docs.docker.com/engine/swarm/key-concepts/
4. https://kubernetes.io/docs/concepts/